QUANTUM TRAFFIC GRIDLOCK

Quantum Optimization for Coordinated Urban Traffic Signals | Technical Report

Scope and status

This report documents the implementation and the available reference-scenario R run. Scenario H results and multi-seed robustness results were not provided with the uploaded outputs, so they are explicitly not claimed here. Update this report after those experiments are run.

1. Problem formulation

The system contains two connected intersections (A and B) over four 30-second slots. Each intersection selects exactly one green phase per slot: 1 = North–South (NS), 0 = East–West (EW). The decision vector is x=(A1,A2,A3,A4,B1,B2,B3,B4), giving eight binary variables and 256 possible schedules.

At each slot, external arrivals and transfers from the previous slot are added to queues. The selected approach serves up to the scenario capacity; the remaining queues contribute to delay and overflow. On the directed A→B link, the model transfers floor(fraction × departures + 0.5) vehicles from the linked approach into B at the beginning of the next slot. Transfers after slot four leave the model. The official objective is delay + overflow_weight × overflow + switch_weight × switches; no switching cost is charged before slot one.

For reference scenario R, capacity=5, storage=7, overflow weight=4, switching weight=0.5, and the A→B link transfers from A's EW approach with fraction 0.6. Arrivals are A(NS)=[3,6,6,3], A(EW)=[5,1,1,2], B(NS)=[3,5,4,3], B(EW)=[2,6,4,6].

2. QUBO construction

The implementation uses a second-order polynomial surrogate: $f(x) \approx c + \Sigma_i h_i x_i + \Sigma_{i<j} J_{ij} x_i x_j$. The constant c is stored as offset; h_i are diagonal entries and J_{ij} upper-triangular entries in Q, so the approximation is $c + x^T Q x$. No auxiliary variables are used. Coefficients are fitted by least squares to exact simulator evaluations on 64 seeded schedules, followed by two active-refinement rounds that add 12 best-predicted unseen schedules each. The output records 88 simulator evaluations for the fit.

This is a scenario-parameterized rule rather than a stored schedule lookup. The quadratic fit approximates min-capacity, overflow-threshold and rounded-transfer behaviour. Its quality must be evaluated against the official cost for all 256 schedules; a good minimizer alone does not establish low surrogate regret.

3. QAOA implementation and reproducibility

The implementation constructs a cost Hamiltonian from the QUBO using x=(1−Z)/2 and applies alternating cost and mixer layers. The uploaded run uses p=3, eight qubits, seeded linear-ramp initialization with small seeded jitter, and COBYLA optimization. The saved metadata reports 299 evaluations, within the 300 limit. The final distribution is sampled for exactly 4,096 shots. OpenQASM 3 circuit, counts, QUBO and decoded schedule are exported separately.

The saved metadata reports internal gate/statevector fidelity of approximately 1.0. This is not an independent Qiskit parse or hardware validation. The mapping documents A slots 1–4 followed by B slots 1–4, with reverse qubit indexing intended to preserve schedule order in measured bitstrings.

4. Reference-scenario result

The uploaded output identifies 11101010 (A=1110, B=1010) as the best schedule observed among 4,096 raw QAOA samples, with official cost 72.0. The saved QUBO exact minimizer is the same schedule and is also scored at 72.0. This establishes agreement for this schedule on R; it does not alone prove global optimality or low regret across all schedules.

Output checks

qubo.json: 8×8 upper-triangular Q, offset, mapping, no auxiliaries, 88 simulator evaluations. circuit.qasm: OpenQASM 3.0, eight qubits and measurements; independent parser check pending. counts.json: 109 observed bitstrings, all 8 bits, total 4,096. schedule.json: A=1110, B=1010, cost 72.0, 299 evaluations.

5. Baselines and evaluation plan

Required comparisons are alternating schedules, per-slot greedy control, exact enumeration, uniform-random expected cost, the official score of the QUBO minimizer, and raw QAOA samples. The supplied diagnostics script includes alternating schedules, enumeration, QUBO-minimizer scoring, random mean and lift/enrichment calculations, but not the specified greedy baseline. No complete baseline table or five-seed summary was supplied.

For each scenario compute C* and C_rand by enumeration. Compute E_Q from 4,096 raw samples, lift=(C_rand−E_Q)/(C_rand−C*), and elite enrichment as the fraction of shots in the 13 lowest-cost schedules divided by 13/256, breaking ties by ascending canonical bitstring. Report at least five fixed seeds and minimum lift clipped to [0,1].

6. Limitations and next steps

Only scenario R outputs were supplied. H and S generalization is unverified. Add the greedy baseline; enumerate all 256 schedules to measure fit error and surrogate regret; independently parse/run the QASM with Qiskit and verify measurement order; run at least five fixed seeds and ablations; and include R and H outputs. Confirm the required command `python run.py --scenario <scenario.json> --seed <int> --out <dir>` in the judging environment.

Current evidence supports a promising R-run artifact set with 4,096 recorded samples. It does not yet support claims of hidden-scenario performance, quantum advantage, or complete competition readiness.

References

Quantum Traffic Gridlock problem statement (QC@IITI / IBM Qiskit Fall Fest 2026); supplied implementation files and R output artifacts.